

Mykhailo Yartsun

INFORMATION ENGINEERING STUDENT · BACKEND, DATA & AI

Hamburg, Germany | +49 1525 8139858 | contact@yartsun.dev | yartsun.dev | github.com/yaamii123

Information Engineering B.Sc. student at HAW Hamburg (6th semester) focused on backend and data systems. Seeking a mandatory internship (Pflichtpraktikum) in software engineering, backend/data engineering, or applied AI/ML as part of the degree program. Hands-on experience building FastAPI/PostgreSQL applications, training ML models, and shipping full-stack AI products.

EDUCATION

HAW Hamburg — B.Sc. Information Engineering | Hamburg, Germany | Oct 2023 – Present

Specialization: Software and Information Technology · Currently in the 6th semester

Kyiv Polytechnic Institute — B.Sc. Software Engineering coursework | Kyiv, Ukraine | Oct 2021 – Nov 2022

Coursework completed until relocation to Germany.

SELECTED PROJECTS

ChestX-Ray AI — Medical Image Analysis | Sole Developer | Jul 2026 | [GitHub](#)

- Fine-tuned ConvNeXt-B on 95,397 NIH ChestX-ray14 training images for 14-label classification, reaching 0.848 macro ROC-AUC on a 16,723-image patient-wise validation split.
- Built a PyTorch/FastAPI inference API with Grad-CAM, three inference modes (fine-tuned ConvNeXt, XRV DenseNet, and ensemble), and a Next.js comparison UI.

FinTrack — Personal Finance App | Primary Backend Contributor, Team Project | May – Jun 2026 | [GitHub](#) | [Live Demo](#)

- Designed FastAPI REST endpoints, JWT authentication, SQLAlchemy models, and PostgreSQL analytics for transactions, budgets, and category reporting.
- Implemented CSV export, spending predictions, and Gemini-powered coaching and report endpoints with multi-key fallback.

AI SQL Assistant | Sole Developer | Jul 2026 | [GitHub](#) | [Live Demo](#)

- Built a multi-user FastAPI/Next.js assistant for SQLite and PostgreSQL that converts natural-language questions into reviewable SQL and streamed results.
- Enforced SELECT-only execution, per-user isolation, JWT authentication, encrypted database URIs, PII redaction, and rate limiting.

Atelier — Visual Fashion Search | Sole Developer | Sep 2026 | [GitHub](#)

- Built visual product search with FastAPI, CLIP embeddings, and PostgreSQL/pgvector HNSW retrieval, followed by hybrid reranking.
- Designed swappable detection, embedding, and ranking providers plus a Next.js interface over a live retailer catalog.

Additional academic work: Java RMI vacuum-robot control system with remote interfaces, serializable DTOs, and UML; normalized PostgreSQL endoscopy database with ER modeling, integrity constraints, and reporting queries.

TECHNICAL SKILLS

Languages: Python, Java, SQL, TypeScript, JavaScript, C#, C++, C

Backend & Data: FastAPI, REST APIs, PostgreSQL, SQLAlchemy, Pydantic, JWT, pgvector

AI & ML: PyTorch, model fine-tuning, Grad-CAM, CLIP, scikit-learn, Pandas, NumPy

Tools & practices: Docker, Git, GitHub Actions, CI/CD, pytest, Playwright, Linux

AI-assisted development: Cursor coding agents for planning, implementation, refactoring, testing, and code review, with test-based verification

LANGUAGES

English: Fluent | **German:** Intermediate | **Ukrainian:** Native | **Russian:** Native